

Stock-AI Platform

UG_HF_03

Andrew Butler, Matthew He, Ethan Roe, Aaron Trelstad

Project Status - Backend

Done

- Portfolio trading, modification, transfers, & soft delete
- Portfolio value tracking (snapshots)

Still working on

- Security refactor (custom -> library)
- Retroactive trading strategy results
- Internal Route for Daily Cron Job

Project Status - Frontend

Done

- Portfolio performance page
- Rename & Delete portfolios
- Transfer assets between portfolios

Still working on

- Email verification
- Password reset

Contributions - Andrew Butler

- Documentation
- Project website
- Additional API endpoints / bug fixes

Project Synopsis

View Edit Delete

Problem

Stock trading is a difficult skill to learn. The stock market is constantly changing and moving, and getting started on smart investments is time consuming and difficult to start without making real investments. Learning platforms exist, but they are expensive services that add on to the already expensive task of trading.

Solution

Our platform, Stock-AI, aims to help beginners learn stock trading completely for free with no risk. It accomplishes this in two ways: first, the Stock-AI platform doesn't charge users. There is no subscription or service cost to the user, and all trades are simulated to remove the risk of financial loss. Second, the platform includes AI integration to allow the user to ask questions and get summaries of relevant information, as well as to get recommendations on any given stock.

Features

Stock-AI has 3 main features that users interact with: the Dashboard, user Portfolios, and Leagues.

Dashboard

The Dashboard serves as the primary way that users get information about the stock market. From here, they can look up any ticker, view its price history, see linked news articles about that company, as well as ask the AI assistant questions and get a recommendation for buy/sell/hold with associated reasoning. Trades can also be performed here, although they must be linked through a singular portfolio the user has. See Figure 1 below for a view of the dashboard.



Stock-AI: AI Integrated Stock Trading Platform

Com S 4020, Team UG_HF_03

Background

This project is the product of Com S 4020 Undergraduate Senior design at Iowa State University (ISU), Spring 2026. For this class, teams of four were created and were assigned projects with clients both within and outside ISU. This specific project was developed under the direction of Hung Phan, a post-doctoral researcher at ISU.

Team Members

Andrew Butler	- Backend development, Client Communication
Matthew He	- Full Stack development
Ethan Roe	- Frontend development
Aaron Trelstad	- Full Stack development, AI Integration

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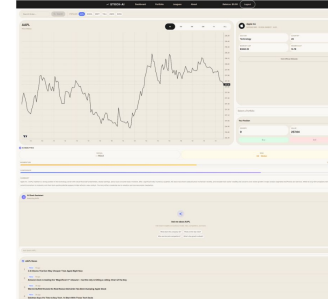
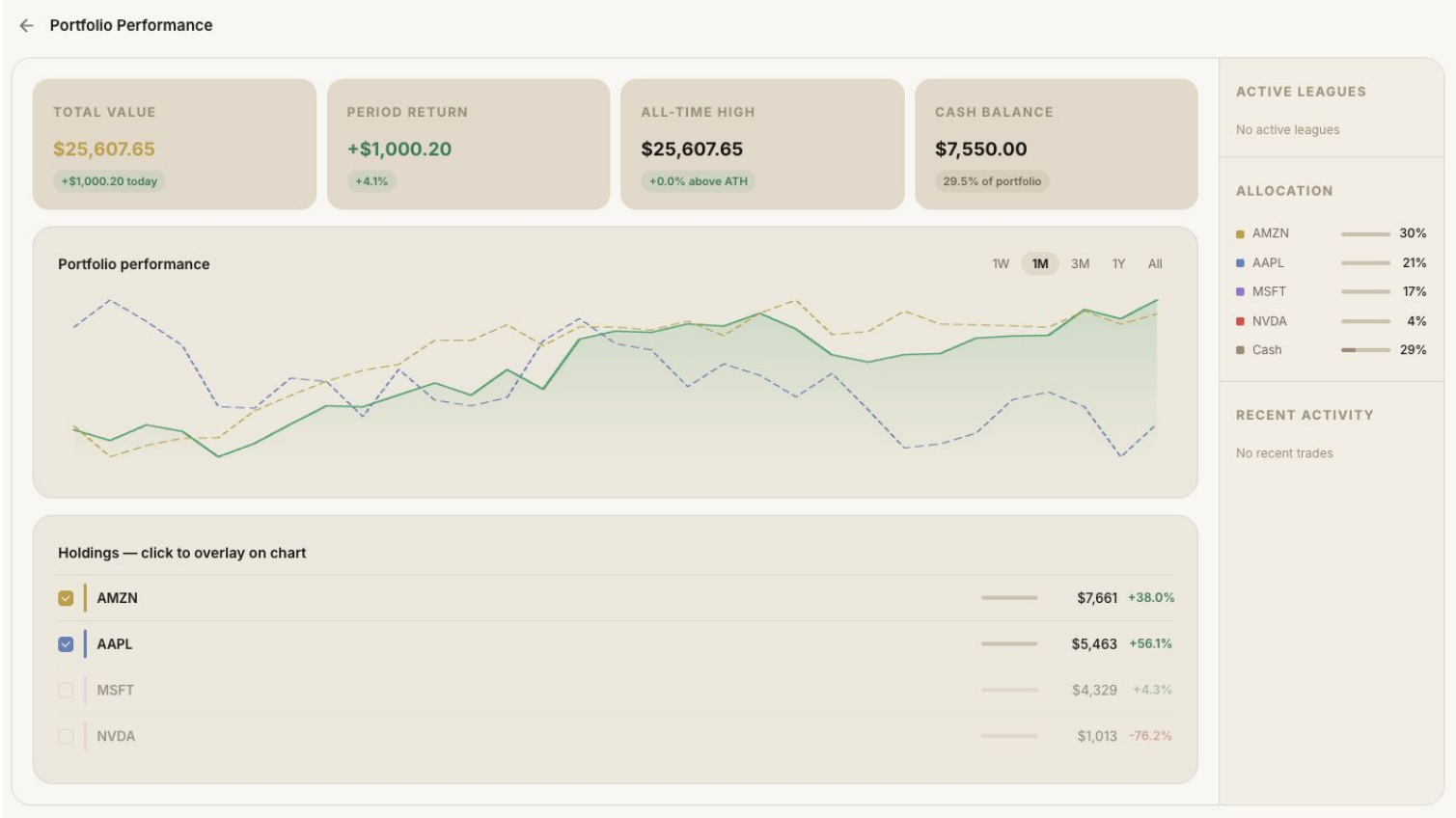


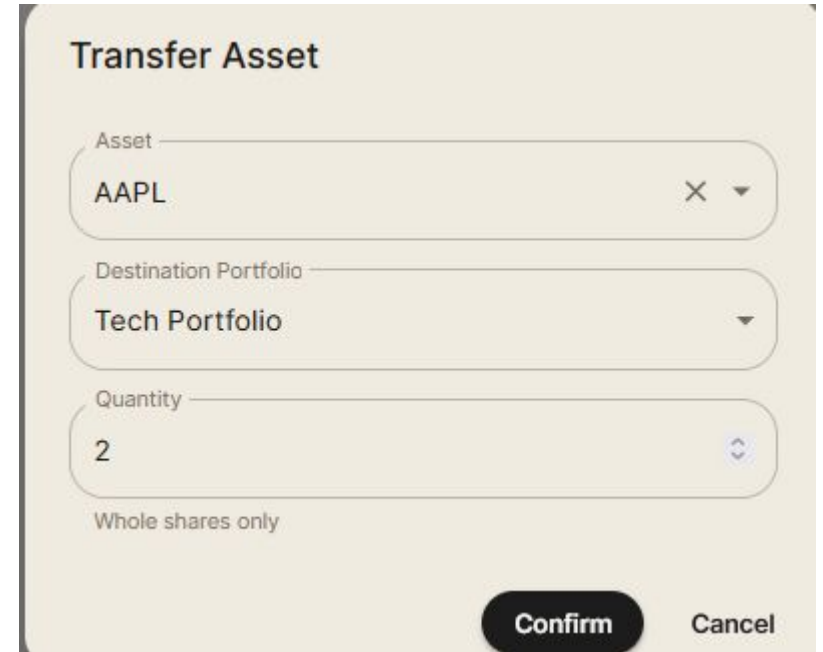
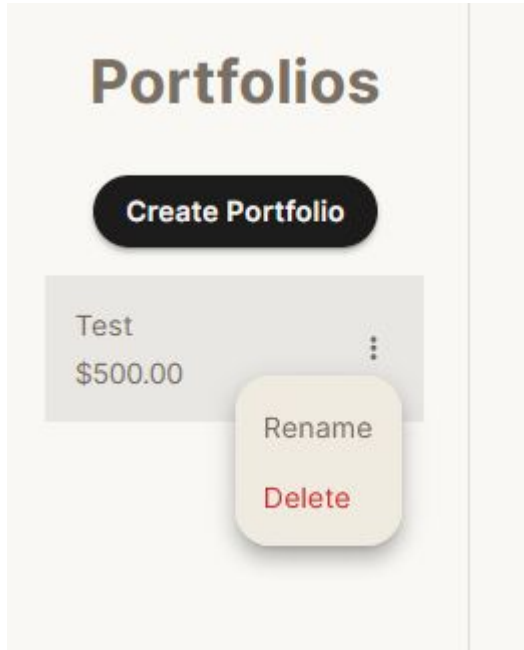
Figure 1: Dashboard page of Stock-AI platform. Note the price graph in the upper left, company information on the upper right, and the AI Chat + Analytical features in the middle.

Contributions - Matthew He

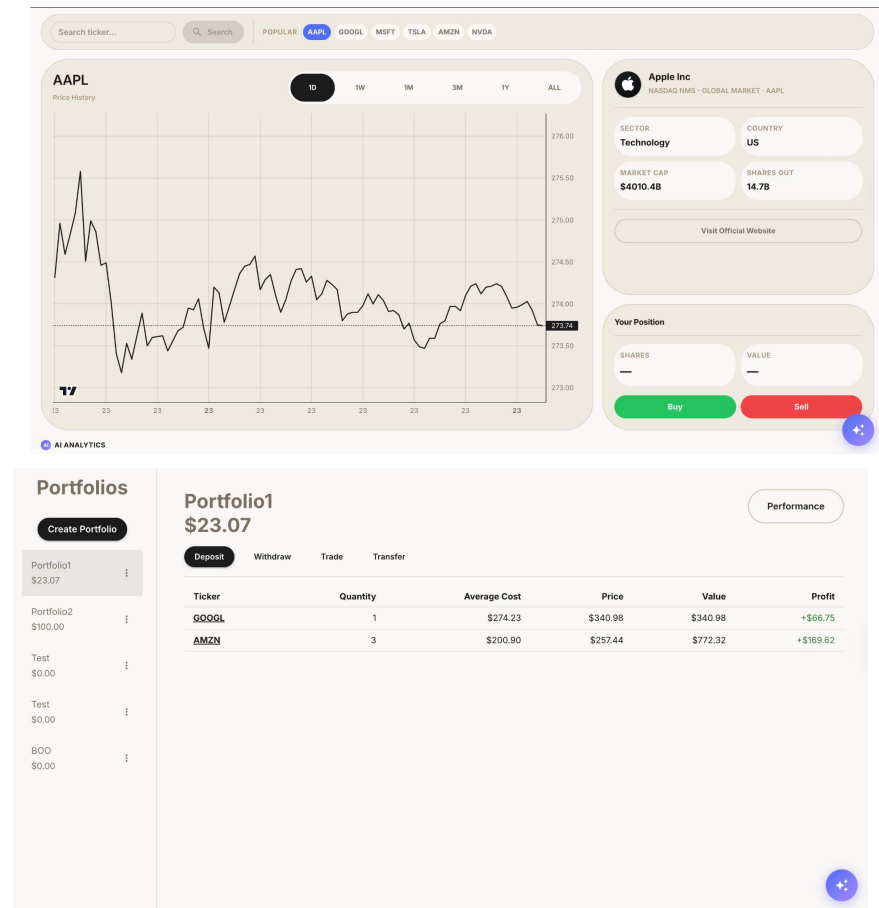
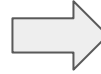
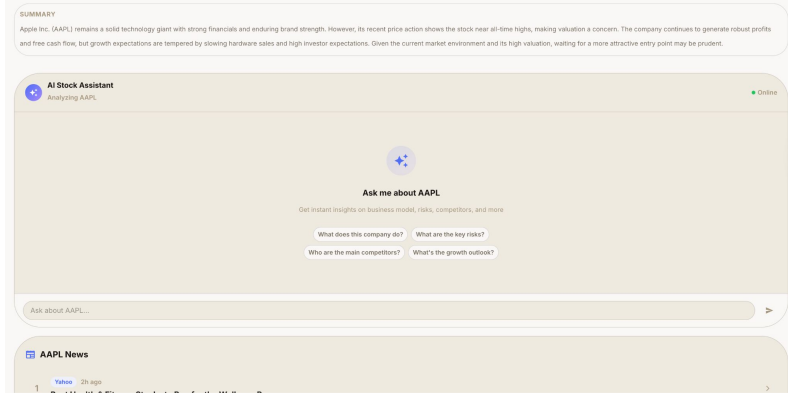


Contributions - Ethan Roe

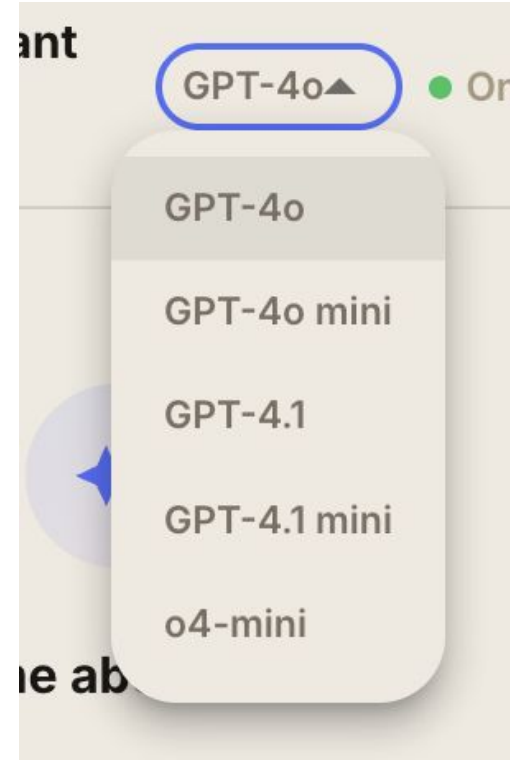
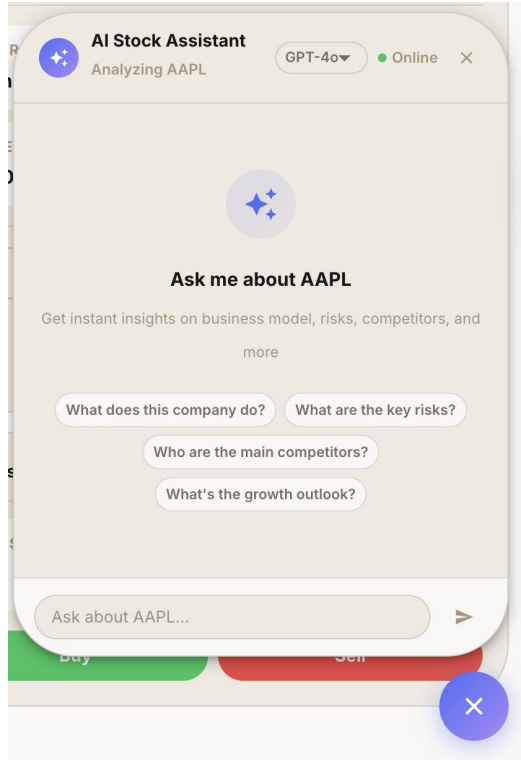
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Contributions - Aaron Trelstad



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